The Limits of Predictions for Online Bipartite Matching

A Unified Experimental Study

Zhuolun Li

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Abstract

Learning-augmented ("with-predictions") algorithms for online bipartite matching have proliferated: MinPredictedDegree consumes a per-node degree predictor, and a family of test-and-fallback schemes tests a type-histogram prediction on a sublinear prefix of the arrivals before committing to follow it or to fall back on an advice-free baseline. These algorithms have been analyzed in isolation. We give the first unified experimental study, placing all of them on a single harness with a common structured prediction-error model, a common optimum, and confidence intervals, across synthetic graphs, six real-world graphs, and real request traces. Our central finding is that on average-case inputs the value of predictions is robustness insurance, not a performance lever: the advice-free baseline is already near-optimal, so the consistency upside of good advice is small, whereas unguarded prediction-following can crash far below the baseline, and the practical worth of the sophisticated algorithms is that they never do. We close with a theoretical outlook: a proved consistency/robustness trade-off inequality, together with a budget–stakes reading of our experiments under which the prefix needed to decide whether to trust advice scales as the inverse square of the advice's upside — a price that, on strong-baseline average-case inputs, exceeds the length of the instance itself. Experiments and outlook deliver one message: on average-case matching, the advice's upside is smaller than the price of finding out whether to trust it.

Statement on the inclusion of published and collaborative work

My dissertation does / does not (delete as appropriate) include work that has been published, submitted or prepared for publication with me as an author. My dissertation does / does not (delete as appropriate) include non-published collaborative work that contains data or contributions from others.

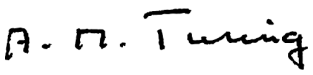
If there are any publications with me as an author and/or any collaborative work that contains data or contributions from others, I have completed the following table.

Chapter	Details	Author Contributions

The information in this table counts as the references to my own publications if they are used as whole chapters or as whole sections of chapters. If work from my own publications have been incorporated in smaller segments, I have included individual references to them in the relevant part of the dissertation in addition to completing the table.

Author's Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes/Research Degree Programmes (delete as appropriate) and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.



Zhuolun Li

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1. Introduction

1.1. Background and motivation

Online bipartite matching is the canonical model of irrevocable sequential allocation. Resources are known in advance; requests arrive one at a time; each must be matched to an available compatible resource — or dropped — immediately, before the rest of the input is seen. It is the abstraction behind online advertising, ride-hailing dispatch, organ exchange, and request routing in modern serving systems. Because decisions cannot be undone, the difficulty lies entirely in committing well under uncertainty about the future.

A recent and influential idea is to give such an algorithm a **prediction** about the input — a hint about which resources will be contended, or how many requests of each kind will arrive — and to ask for two guarantees at once: **consistency**, meaning near-optimal performance when the prediction is good, and **robustness**, meaning no worse than a prediction-free algorithm when the prediction is wrong. For online matching specifically, two families of such *learning-augmented* algorithms have appeared: MinPredictedDegree, which consumes a per-resource degree prediction, and a family of *test-and-fallback* schemes, which test a request-type prediction on a short prefix of arrivals before deciding whether to trust it.

These algorithms come with attractive worst-case guarantees, yet — as this thesis demonstrates experimentally — their average-case behavior is strikingly muted: on realistic workloads the sophisticated predictions rarely *help*, while the algorithms' real

value is that they do not *hurt*. This gap between the worst-case promise and the average-case reality is the puzzle that motivates this thesis. The algorithms had, moreover, only ever been studied *in isolation* — each on its own input model, against its own notion of prediction error, and largely through theory — so there was no common ground on which to compare them, quantify how much a prediction actually buys, or explain the gap. This thesis provides that common ground and, ultimately, an explanation.

1.2. Research objectives

The thesis is organized around three questions:

1. **How do the learning-augmented online-matching algorithms actually compare**, head to head, under a single prediction-error model and on common inputs — and how much does a prediction help, at what cost, on realistic data?
2. **What are the failure modes of the adaptive test-and-fallback mechanism** — the cost of its test, the calibration of its accept/reject decision, and where it goes wrong?
3. **Why does the average-case experience differ so sharply from the worst-case promise** — is the observed wall an artifact of particular algorithms and generators, or is it *necessary*?

The first two are answered experimentally; the third, theoretically.

1.3. Contributions

- **A unified experimental benchmark** (Chapter 4) that places the advice-free baselines, MinPredictedDegree and its augmentations, and the test-and-fallback algorithms on one harness — common graphs, a common structured prediction-

error model, a common optimum, and confidence intervals — the first head-to-head comparison of these families.

- **A characterization of predictions as robustness insurance** (Chapters 4, 7): unguarded prediction-following crashes *below* the advice-free baseline under bad predictions; two distinct mechanisms — structural (the augmentations) and adaptive (test-and-fallback) — restore safety with different consistency/robustness trade-offs; and the consistency upside is small on average-case inputs, so the value is downside protection. The picture is validated on synthetic graphs, six real-world graphs, and real request traces, including with a cheap, non-ML historical predictor.

- **An empirical engagement with the order-error theory** (Chapter 5): MinPredictedDegree's loss is governed by a Kendall-τ order error onto which several error models collapse. We credit Aamand–Chen–Indyk's Appendix D for the order-dependence itself and contribute a characterization of the known bound's looseness and saturation and of the right order measure — not a rediscovery that order matters.

- **The first empirical study of test-and-fallback** (Chapter 6): its testing cost, a threshold-calibration pathology in which a *more accurate* test yields a *worse* decision, its recalibration and the resolution limit that recalibration exposes, and a benchmark of the dynamic combiner revealing an irrevocability penalty that explains why matching requires *test-then-commit*.

- **A quantified account of why the wall stands** (Chapter 6 and §10.2): the resolution-limit finding — no practical acceptance threshold can capture an upside that sits below its estimator's noise — is shown to persist across the whole difficulty range (Figure 6.4) and is given a theoretical outlook in the conclusion: a proved consistency/robustness trade-off inequality, and a *budget–stakes* reading under

which the prefix needed to decide scales as the inverse square of the advice's upside — a price that exceeds the instance length exactly where the baseline is strong. The full formal development is deliberately deferred to companion work in preparation, and the thesis claims no theorem beyond the stated inequality.

Alongside these, the thesis documents (Chapter 8) the exploratory directions it pursued and the negative results they returned — an attempt to *learn* the predictor for extra performance, and an attempt to find a serving regime where predictions genuinely help — both of which reinforce the central finding; the question they raise — is the wall forced? — is taken up in the concluding outlook (§10.2).

1.4. Thesis organization

Chapter 2 surveys the background: online matching and its input models, the learning-augmented paradigm, the specific prediction-based matching algorithms, and the distribution-testing results behind their tests. Chapter 3 fixes the model and methodology and validates the experimental harness by reproducing a subset of the Borodin et al. study. Chapter 4 presents the unified benchmark and its four findings. Chapter 5 examines what governs the (small) loss — order error — and its relation to the known bound. Chapter 6 studies the test-and-fallback mechanism in depth. Chapter 7 tests external validity on real predictors and real graphs. Chapter 8 reports the exploratory directions and negative results. Chapter 9 presents the AI-inference serving case study. Chapter 10 concludes: it summarizes the findings, gives a theoretical outlook on why the wall is forced (§10.2), and discusses limitations and future work. The recurring thesis, established experimentally and then explained in outlook, is a single sentence: **on average-case online matching, predictions are robustness insurance rather than a performance lever — and the upside they offer is smaller than the price of finding out whether to trust them.**

2. Background and Related Work

This chapter sets up the technical context the thesis builds on: the online bipartite-matching problem and its input models (§2.1), the known-i.i.d. model we work in (§2.2), the learning-augmented ("with-predictions") paradigm and its consistency/robustness language (§2.3), the specific prediction-based matching algorithms we benchmark (§2.4), the distribution-testing results behind the algorithms' tests and the outlook of §10.2 (§2.5), and the gaps in the literature this thesis addresses (§2.6).

2.1. Online bipartite matching and competitive analysis

In online bipartite matching, one side of a bipartite graph — the *offline* resources — is known in advance, while the vertices of the other side — the *online* requests — arrive one at a time. On each arrival the algorithm sees the request's incident edges and must either match it to a currently unmatched neighbor or leave it unmatched, **immediately and irrevocably**. The goal is to maximize the size (or, in weighted variants, the value) of the matching. Performance is measured by the **competitive ratio**, the (expected) ratio between the algorithm's matching and an offline optimum on the same input.

The difficulty of the problem depends entirely on the assumed *input model*:

- **Adversarial arrival order.** The requests and their order are chosen by an adversary. The seminal result of Karp, Vazirani and Vazirani [11] is that the randomized **Ranking** algorithm — fix a uniformly random priority order on

the resources and match each request to its highest-priority available neighbor — achieves competitive ratio $1 - 1/e \approx 0.632$, and that this is optimal for the adversarial model. Deterministic algorithms cannot beat 1/2.

- **Random arrival order.** The graph is adversarial but the requests arrive in a uniformly random order; this is strictly easier than adversarial order and admits ratios above $1 - 1/e$.
- **Known-i.i.d.** Requests are drawn independently from a *known* distribution over request types (§2.2); this is the average-case model and the one this thesis studies.

Because the known-i.i.d. and random-order models are more benign than adversarial order — formally, Known-I.I.D. $\leq$ Random-Order in difficulty — algorithms and guarantees designed for the harder models carry over, a fact we use throughout.

2.2. The known-i.i.d. model

In the known-i.i.d. model there is a bipartite *type graph*: each online **type** ℓ has a fixed neighborhood $N(\ell)$ among the offline resources, and an instance is generated by drawing m arrivals independently from a known distribution p over types. The type graph and p are known; the realized sequence is not. This models settings — ad serving, recurring request streams — where the population of requests is statistically stable even though individual arrivals are not.

A line of work has pushed the worst-case (over type graphs) competitive ratio above the $1 - 1/e$ adversarial barrier: Feldman et al. [7] first beat it (0.67) via a flow-based *blue/red* decomposition of a suggested matching; Manshadi, Oveis Gharan and Saberi [13] and Jaillet–Lu [9] improved the ratio (to ≈ 0.702 and ≈ 0.729 respectively) using LP-based and list-based online policies; subsequent work refined it further. These are the "sophisticated" algorithms whose worst-case optimality motivates their use.

Crucially for this thesis, Borodin, Karavasilis and Pankratov [2] conducted an experimental study of these algorithms and found that on *average-case* i.i.d. instances the picture is very different from the worst case: the simple algorithms (Greedy, Ranking) perform almost as well as the sophisticated ones, whose advantage is a worst-case, not a typical-case, phenomenon. This empirical observation — that on typical inputs the simple baseline is already near-optimal — is the seed of the thesis's central finding, and we reproduce a subset of it as validation (Chapter 3).

2.3. Learning-augmented algorithms

The **algorithms-with-predictions** (or *learning-augmented*) paradigm, surveyed in [14], equips an online algorithm with a prediction about the unknown input and asks for two guarantees simultaneously:

- **Consistency:** near-optimal performance when the prediction is accurate;
- **Robustness:** performance no worse than a prediction-free guarantee when the prediction is arbitrarily wrong.

The paradigm was crystallized by Lykouris and Vassilvitskii [12] for competitive caching, who showed how to interpolate between trusting and ignoring a predictor while bounding both consistency and robustness. A large literature followed across ski rental, scheduling, and other online problems, including the *optimal* consistency/robustness trade-off analyses of Wei and Zhang [20], which establish problem-intrinsic Pareto frontiers between the two objectives. A recurring theme is that robustness is obtained by *hedging* — combining the predictor's advice with a safe default so that a bad prediction cannot cause catastrophe. The blind-follow-with-switching **combiner** of Chłędowski, Polak, Szabucki and Żołna [5], introduced in an experimental study of robust learning-augmented caching, is one such hedging mechanism; we port and benchmark it (Chapter 6). That paper

is also the closest methodological template for this thesis's experimental half. Recent theoretical work further analyzes how the achievable ratio degrades *continuously* with prediction accuracy in a distributionally-robust formulation [21], complementing the discrete consistency/robustness endpoints used here.

2.4. Predictions for online bipartite matching

Two strands apply the with-predictions paradigm to online matching, differing in the *prediction object* they consume.

Degree predictions: MinPredictedDegree. Aamand, Chen and Indyk [1] propose **MinPredictedDegree (MPD)**: given a prediction μ of each offline resource's degree (how contended it will be), match each arrival to its available neighbor of *minimum predicted degree* — i.e. protect the resources predicted to be rarest. MPD is robust by construction: a constant (useless) predictor reduces it to Ranking. A key structural fact, which the thesis engages in Chapter 5, is that MPD depends on μ *only through the order it induces*; the authors' Appendix D bounds the matching loss by an order quantity ($n - \mathrm{LIS}$, the number of resources not in the longest non-decreasing subsequence of the true degrees ordered by μ), and shows in particular that a monotone (order-preserving) prediction incurs zero loss.

Type-histogram advice and test-and-fallback. A second strand takes the prediction to be a *histogram* $\hat{c}$ over request types (how many of each type will arrive). Choo et al. [6] introduce **TestAndMatch**: build a matching from $\hat{c}$ and Mimic it, but first spend a sublinear prefix of arrivals *testing* whether the observed type frequencies match $\hat{c}$ — using an ℓ_1-distance tester adapted from Jiao, Han and Weissman [10] — and fall back to Ranking if they do not. Burathep, Erlebach and Moses [3] generalize this ("Test-and-Match+") to the random-arrival model and to imperfect knowledge of the matching size. Both papers give *upper* bounds (algorithms); their only lower bound (Choo

et al.'s Theorem 3.1) is a generic *adversarial* indistinguishability result (no algorithm is 1-consistent and $> \frac{1}{2}$-robust), and neither proves a lower bound in the stochastic model. Notably, Choo et al.'s acceptance threshold already *couples* to the baseline competitive ratio β — but constructively, inside the algorithm design, not as a lower bound. What that coupling costs — how large a prefix the follow/fallback decision fundamentally requires — is a question this thesis engages empirically (Chapter 6) and in a theoretical outlook (§10.2); the full formal development is deferred to companion work.

2.5. Distribution testing

The test at the heart of the algorithms above is a question in **distribution testing**: given samples from an unknown distribution p over a support of size r and a known reference q, decide how far p is from q in ℓ_1 (total-variation) distance. Two regimes must be distinguished:

- **Identity (non-tolerant) testing** — distinguish $p = q$ from $\|p - q\|_1 \geq \varepsilon$ — has sample complexity $\Theta(\sqrt{r}/\varepsilon^2)$ [16, 19], *sublinear* in the support size.
- **Tolerant testing / distance estimation** — distinguish $\|p - q\|_1 \leq \varepsilon_1$ from $\geq \varepsilon_2$ for $0 < \varepsilon_1 < \varepsilon_2$, i.e. estimate the distance rather than test equality — is provably *much harder*: Valiant and Valiant [18] showed it requires $\Theta(r/\log r)$ samples, *near-linear* in the support. Jiao, Han and Weissman [10] gave matching bounds for ℓ_1-distance estimation, and Canonne, Jain, Kamath and Li [4] precisely characterized the whole $(\varepsilon_1, \varepsilon_2)$ landscape, showing that for constant tolerances the cost jumps to the "barely sublinear" $\tilde{\Theta}(r/\log r)$.

The near-quadratic gap between $\sqrt{r}$ (testing) and $r/\log r$ (tolerant testing) matters twice in this thesis. It is why the deployable versions of TestAndMatch fall back to an empirical surrogate for their tester, and it is why that surrogate is blind on large supports

— the resolution limit measured in Chapter 6. Whether the barrier also dooms every *other* decision rule turns out to be subtler than it first appears; the concluding outlook (§10.2) returns to this point.

2.6. Positioning of this thesis

Against this background, three gaps stand out, which the thesis addresses in turn:

1. **No unified empirical comparison.** The matching algorithms above were each studied in isolation, on their own input families and error models, and largely in theory; there is no head-to-head experimental benchmark under a common harness. (Chapters 4–7.)
2. **No empirical study of test-and-fallback.** The distribution test at the heart of [6, 3] has no deployable implementation (the authors themselves fall back to an empirical surrogate), and its testing cost, threshold calibration, and failure modes have not been measured. (Chapter 6.)
3. **No quantification of what the prefix test costs.** No prior work measures — or bounds — how large a prefix the follow/fallback decision requires on strong-baseline instances, where the upside to be captured is smallest. (Chapter 6 empirically; §10.2 in outlook.)

The thesis closes the first two gaps experimentally and takes a first, quantified step at the third — an empirical resolution limit plus a theoretical outlook — arriving at a single unifying statement: *on average-case online matching, predictions are robustness insurance rather than a performance lever*, supported by experiments across synthetic graphs, real graphs, and real traces.

3. Model, Algorithms, and Methodology

Chapter 2 placed the problem in its field. This chapter fixes the precise model and notation used throughout (§3.1), details the algorithms as we implement them (§3.2), the prediction objects and structured error models (§3.3), the consistency/robustness measures (§3.4), and the experimental harness (§3.5). It closes by validating the harness against the published Borodin et al. figures before any contribution is built on it (§3.6).

3.1. The known-i.i.d. model, precisely

There is a set R of n offline **resources** and a **type graph** on r online **types**; type ℓ has neighborhood $N(\ell) \subseteq R$. An instance is generated by drawing m arrivals i.i.d. from a distribution p over the r types; the i-th arrival reveals its type (hence $N(\cdot)$) and must be matched to a currently unmatched resource in its neighborhood, immediately and irrevocably, or left unmatched. The type graph and p are known to the algorithm; the realized arrival sequence is not. Following the standard *integral-types* convention, the default is $m = n$ with uniform p, so each type's expected count is an integer; the classical setting has $r = n$ (each offline vertex a distinct type), and the *few-types* setting used for the histogram-advice experiments has $r \ll n$.

The performance measure is the **competitive ratio** $\rho(\mathrm{ALG}) = \mathbb{E}[|\mathrm{ALG}|]/\mathbb{E}[|\mathrm{OPT}|]$, where OPT is a maximum matching of the *realized* instance, computed exactly by Hopcroft–Karp [8]. The advice-free baseline throughout is KVV Ranking, whose ratio

we write ρ_{base} (the **baseline strength**).

3.2. Algorithms

All greedy-type algorithms are instances of one primitive: given a total order (**rank**) on the resources, **GreedyWithPermutation** matches each arrival to its available neighbor of minimum rank. This makes the algorithm family a single code path parameterized by the rank, which is important both for a fair comparison and for the theory.

- **Advice-free baselines.** SimpleGreedy (identity rank); **Ranking** (uniformly random rank; the baseline); and the stochastic-matching algorithms Feldman and Jaillet–Lu, which precompute a suggested matching by max-flow — Feldman via a cap-$\{2, 1, 2\}$ flow network and a blue/red decomposition of the unit-flow subgraph, Jaillet–Lu via a cap-$\{3, 2, 3\}$ network yielding a fractional solution $f^\star \in \{0, \frac{1}{3}, \frac{2}{3}\}$ and per-type list distributions. Each has a *non-greedy* variant (follow the suggestion only) and a *greedy* variant (fall back to any available neighbor).

- **Degree predictions.** MinPredictedDegree (MPD) is GreedyWithPermutation ranked by ascending predicted degree $\mu \in \mathbb{R}^R$. Constant μ gives Ranking; the true realized degrees give the consistency ceiling (MinDegree). The **augmentations** Feldman(MPD) and JailletLu(MPD) use the MPD rank only as the tie-break of the greedy fallback, so the worst-case-optimal base matching carries the load.

- **Type-histogram advice and test-and-fallback.** From a count vector $\hat{c}$ over types we build an advice matching $\hat{M}$ (a maximum matching of $\hat{c}$ copies) and record its per-type partners. **FollowPrediction** *mimics* $\hat{M}$ — routes every arrival to its type's advice partner; **TestAndMatch** (Choo and BEM variants) *mimics* over a length-k prefix, tests the empirical ℓ_1 distance between the prefix type frequencies and $q = \hat{c}/\|\hat{c}\|_1$ against a threshold τ, then continues or falls back to Ranking. We

also port the Chłędowski-style dynamic **combiner** as a benchmarked baseline (not a contribution).

3.3. Prediction objects and structured error models

The families consume different prediction objects — degree vectors μ and type histograms $\hat{c}$ — so each is corrupted by its own knob and reported in a parallel panel. For μ we use four structured error models, each returning both an ℓ_1 *magnitude* and a normalized **Kendall-τ order error**: random-flip, systematic-bias (a monotone rescale — order-preserving, hence Kendall-$\tau \equiv 0$ by construction), adversarial (reflection of a fraction of entries), and distribution-drift (blend toward an independently drawn graph's degrees). For $\hat{c}$ we blend the realized counts toward a concentrated random target by a fraction $\eta \in [0, 1]$, reporting the induced $\ell_1(p, q)$. Because MPD depends on μ only through the induced order, the order/magnitude distinction is the axis of Chapter 5.

3.4. Consistency and robustness

For a prediction-consuming algorithm, **consistency** is its ratio under perfect advice and **robustness** is its worst-case ratio under adversarial advice. A useful algorithm is consistent *and* never (much) below ρ_{base}; the tension between the two is the subject of Chapter 6 and of the outlook in §10.2.

3.5. The experimental harness

The harness is a small, dependency-light Python codebase (NumPy, SciPy, NetworkX). All randomness comes from independent, reproducible streams obtained by NumPy's spawn mechanism — by default four streams (graph, instance, Ranking, Jaillet–Lu), with additional streams spawned for prediction generation and perturbation, so adding an

algorithm never disturbs existing results. We use **paired trials**: within a comparison, every algorithm and error level reuses the same graph, arrival sequence, OPT, and tie-break seed, so differences are attributable to the prediction alone. Reported ratios are means over trials with 95% normal-approximation confidence intervals. Every figure and table in the thesis is regenerated from a fixed seed by a single script (Appendix A).

3.6. Fidelity check: reproducing Borodin et al.

Before building prediction-based algorithms on the harness, we validated it against the published experimental study of Borodin, Karavasilis and Pankratov [2]. This is a **fidelity check, not a contribution**: its purpose is to confirm that the generators, the i.i.d. sampler, the Hopcroft–Karp optimum, and the algorithm implementations reproduce the paper's qualitative findings, so that later differences can be attributed to the algorithms rather than to infrastructure. Our stack (Python/NetworkX) differs from the paper's (C++/Edmonds–Karp), so we target *qualitative* agreement, accepting small absolute differences (≤ 0.02).

We reproduced the core four algorithms (six greedy/non-greedy variants) on two random graph families at $n = 1000$, $m = n$, 100 trials per parameter value (matching the paper's setup), under a single master seed.

Erdős–Rényi (edge probability c/n; the paper's Fig. 9, partial). Sweeping c reproduces the characteristic U-shape and its hard case: SimpleGreedy attains its minimum 0.864 at $c \approx 4.9$, and the greedy variants of the complex algorithms their minima ≈ 0.884 at $c \approx 5.3$. Ranking is indistinguishable from SimpleGreedy (maximum difference 0.0017 across all 75 values — the paper omits Ranking's curve for exactly this reason), and the non-greedy variants degrade monotonically as c grows ($0.987 \to 0.729$ for Feldman, $0.985 \to 0.764$ for Jaillet–Lu). Every one of the paper's five prose claims we checked holds.

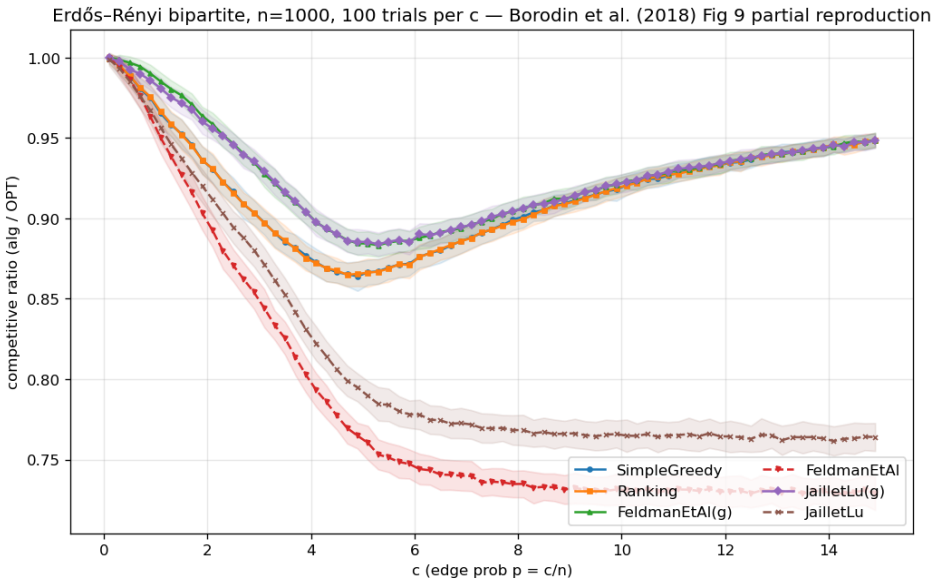


Figure 3.1.: Erdős–Rényi reproduction: the characteristic U-shape, greedy minima near $c \approx 4.9$, non-greedy degradation as c grows.

Random left-regular (left-degree d; the paper's Fig. 18, partial). The pattern repeats with the hard case at $d = 5$ (SimpleGreedy minimum 0.890), Ranking $\approx$ SimpleGreedy (max difference 0.005), non-greedy degradation as d grows, and greedy complex variants $\approx$ SimpleGreedy asymptotically.

A cross-family observation. Combining the two sweeps surfaces a non-obvious fact: the non-greedy Feldman and Jaillet–Lu algorithms converge to the *same* asymptotic ratio in *both* families — 0.729 and 0.764 respectively, within 0.002 across families — and both sit *above* their worst-case theoretical bounds (0.670 and 0.729) by +0.06 and +0.03. This hints at a universal "average-case asymptotic constant" distinct from the worst-case guarantee — an early, concrete instance of the thesis's recurring theme that the average case is far more benign than the worst case, which the prediction-error sweeps of later chapters probe in earnest.

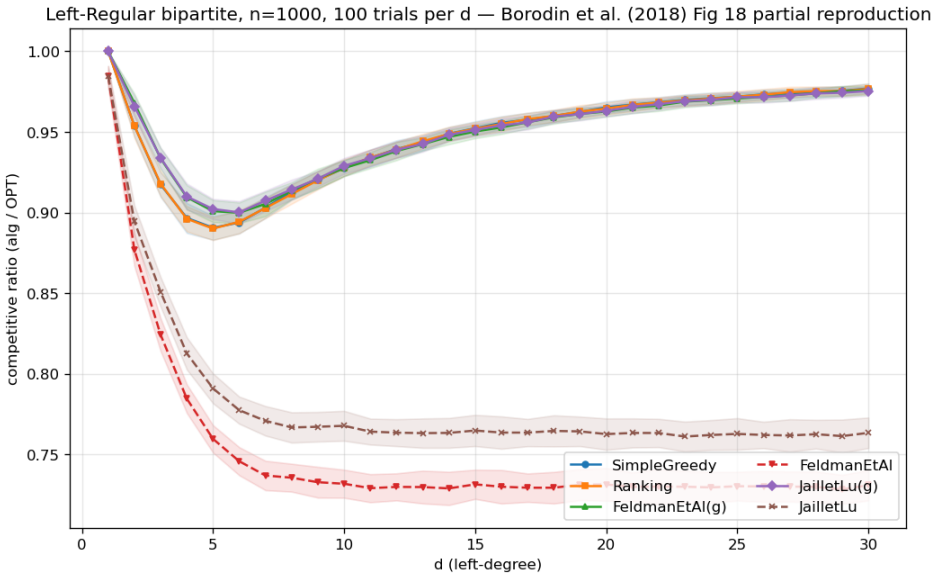


Figure 3.2.: Random left-regular reproduction: the hard case at $d = 5$, Greedy $\approx$ Ranking.

Verdict. The harness reproduces the published qualitative findings on both families, including the locations of the hard cases and the near-identity of Greedy and Ranking. The foundation is trusted; the contributions of Chapters 4–9 are built on it. (Full reproduction tables and the paper-claim checklist are in Appendix A.)

4. The Unified Benchmark

Having validated the harness (Chapter 3), we now place all three algorithm families on it and establish the thesis's organizing finding. This chapter reports competitive ratios (mean ± 95% CI) as a function of prediction quality, in three panels chosen to span the regimes each family was designed for; the four findings it draws out (§4.2) recur through the rest of the thesis.

4.1. Design

The families consume different prediction objects, so each is driven by its own corruption knob and reported in a parallel panel (Chapter 3); the shared harness — graphs, OPT, CI methodology, and the advice-free floor — is what makes the panels comparable.

Instance format and notation. Every panel uses the instance format of Chapter 3: n offline resources, r online request types, and m arrivals drawn i.i.d. from the type distribution; throughout this chapter $m = n$, so requests and resources are balanced. The panels differ *only* in the type graph connecting requests to resources — that single change is what moves the input between the regimes the two prediction families were designed for:

- **Panel A — clvb_zipf** ($n = 1000$, 60 trials): resource degrees follow a Zipf power law with exponent 1.0 — a few resources are heavily contended while most are

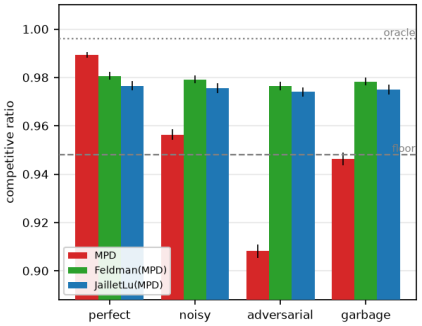
rarely eligible. This heavy-tailed profile gives a *degree* predictor genuine signal to carry.

- **Panel B — left-regular** d=5 ($n = 1000$, 60 trials): each arriving request connects to exactly $d = 5$ uniformly random resources, so resource degrees are nearly homogeneous — the hard case of Chapter 3, where a degree predictor has almost no signal left to carry.
- **Panel C — few-types** r=8 ($n = 2000$, 50 trials): only $r = 8$ distinct request types, each arriving $\approx n/r = 250$ times on average — the near-perfect-matchable, few-types regime the *histogram*-advice algorithms are calibrated for; their test-and-fallback test inspects a prefix of $k = 200$ arrivals.

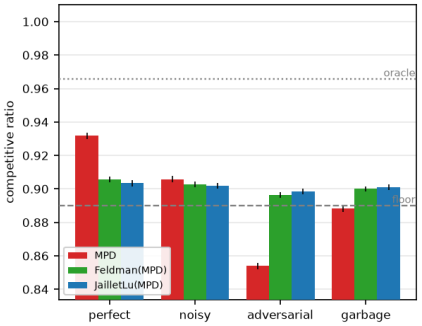
Panels A and B thus exercise the degree-prediction family (MPD and its augmentations); Panel C exercises the histogram-advice family (FollowPrediction, TestAndMatch, and the combiner).

Shared methodology. Every panel runs with *paired trials*: within a panel, every algorithm and quality level reuses the same graphs, arrival sequences, realized optima (Hopcroft–Karp), and tie-break seeds, so column differences are attributable to the prediction alone. All randomness derives from four independent streams (graph, instance, algorithm tie-breaks, prediction perturbation) spawned from one master seed per panel. Entries are means over trials with 95% normal-approximation confidence intervals, tight throughout (± 0.001–0.003). The quality columns instantiate the error models of §3.3 — degree panels: *perfect* (true realized degrees), *noisy* (random-flip at strength $\frac{1}{2}$), *adversarial* (order-reversing reflection), *garbage* (independent random μ, $\equiv$ Ranking); advice panel: the true histogram blended toward a concentrated random target by $\eta \in \{0, 0.3, 0.6, 1.0\}$ (*perfect / mild / bad / garbage*). **Table 4.1** presents each panel's ratios beside its bar chart; the findings follow in §4.2.

Panel A — clvb_zipf	perfect	noisy	advers.	garbage
Ranking (floor)	0.948	—	—	—
MinDegree (oracle)	0.996	—	—	—
MPD	0.989	0.956	**0.908**	0.946
Feldman(MPD)	0.981	0.979	0.976	0.978
JailletLu(MPD)	0.977	0.976	0.974	0.975
Feldman (base)	0.887	—	—	—
JailletLu (base)	0.901	—	—	—



Panel B — left-regular d=5	perfect	noisy	advers.	garbage
Greedy = Ranking (floor)	0.890	—	—	—
MinDegree (oracle)	0.966	—	—	—
MPD	0.932	0.906	**0.854**	0.888
Feldman(MPD)	0.906	0.903	0.896	0.900
JailletLu(MPD)	0.903	0.902	0.898	0.901
Feldman (base)	0.758	—	—	—
JailletLu (base)	0.789	—	—	—



Panel C — few-types r=8	perfect	mild	bad	garbage
Ranking (floor)	0.990	—	—	—
MinDegree (oracle)	0.999	—	—	—
FollowPrediction	1.000	0.832	0.679	**0.472**
TestAndMatch (Choo)	1.000	0.984	0.989	0.990
TestAndMatch (BEM)	0.998	0.988	0.988	0.968
Combiner *(benchmark)*	0.990	0.990	0.990	0.990

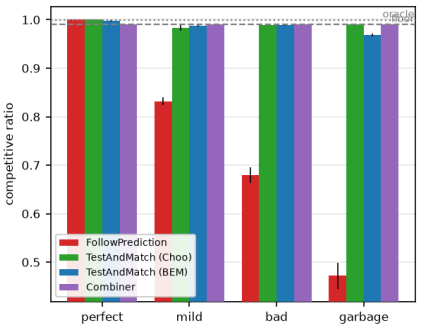


Table 4.1.: The unified benchmark. Each panel's competitive ratios (means over paired trials; every 95% CI ≤ 0.003) sit beside their bar chart (error bars: 95% CIs; dashed line: advice-free floor; dotted: oracle ceiling). Bold marks dips below the floor.

4.2. Four findings

(F1) Robustness is engineered, not free: naive followers crash below the floor. Both unguarded prediction-followers dive under the advice-free Ranking floor once the prediction is adversarial or garbage: MPD falls to $0.908 < 0.948$ (Panel A) and $0.854 < 0.890$ (Panel B), and FollowPrediction collapses to $0.472 \ll 0.990$ (Panel C). Using either *unguarded* is strictly worse than using no prediction at all. Every robust algorithm — the augmentations, TestAndMatch, the combiner — avoids this by construction.

(F2) Two distinct robustness mechanisms, with different shapes. *Structural* robustness (Feldman(MPD), JailletLu(MPD)): the worst-case-optimal base matching carries the load and the prediction only breaks ties, so performance is nearly *flat* — Feldman(MPD) moves only $0.981 \to 0.976$ from perfect to adversarial (Panel A). It cannot crash but caps the upside (never reaching the 0.996 oracle). *Adaptive* robustness (TestAndMatch): test a sublinear prefix, then commit — capturing the upside when advice is good (Choo 1.000) and holding the floor when it is bad (0.990). On Panel C it is the only algorithm on the upper envelope at both ends. The two mechanisms trade consistency for robustness in opposite ways; **Figure 4.1** plots every algorithm on the consistency–robustness plane, where the opposite trades — and the empty region beyond TestAndMatch toward the ideal top-right corner — are visible at a glance.

(F3) The consistency upside is small on average-case inputs; the spread lives on the bad-advice side. On few-types the advice-free Ranking is already 0.990 and MPD-with-true-degrees is 0.999 — under 0.01 for any advice to add on the good side. Every wide gap in Panel C is a *downside* gap. This is the thesis in one panel; Chapter 6 shows why no affordable test escapes it, and the concluding outlook (§10.2) why it is forced.

(F4) The augmentation rescues structurally weak base algorithms. Feldman

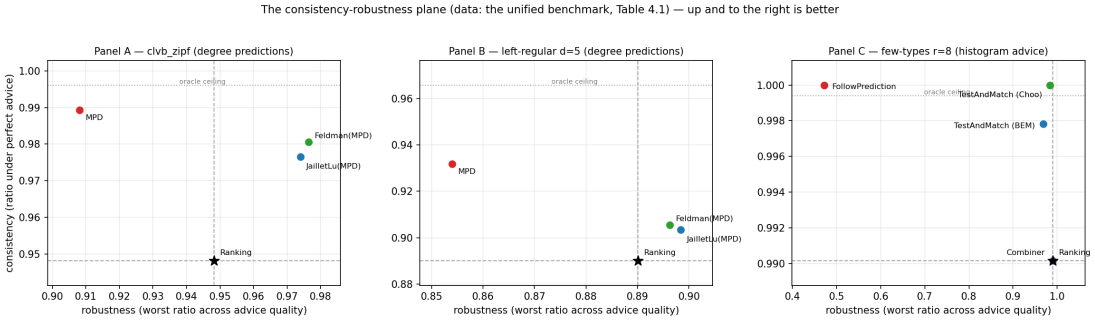


Figure 4.1.: The consistency–robustness plane (the data of Table 4.1); dashed lines mark the advice-free floor, dotted the oracle ceiling. TestAndMatch sits nearest the ideal top-right corner.

and Jaillet–Lu are tuned for the worst-case ratio and are the *weakest* advice-free entries on these average-case inputs (Panel B: 0.758 / 0.789, below Greedy's 0.890); the MPD augmentation lifts them to ≈ 0.90. The prediction does *more* for the worst-case-designed algorithms than for greedy — a pairing visible only under a unified table.

4.3. Chapter summary

On average-case matching the advice-free baseline is already near-optimal, unguarded prediction-following is unsafe, and the value of the sophisticated algorithms is downside protection delivered by one of two mechanisms. Chapters 5–7 sharpen each part — what governs the (small) loss, what the adaptive test costs, and whether the picture survives on real data — and the concluding outlook (§10.2) explains why the wall is forced, not accidental.

5. What Governs the Loss: Order Error and ACI's Bound

Chapter 4 showed that on average-case inputs the loss of a degree-prediction algorithm is small. This chapter asks what *governs* that loss. MinPredictedDegree matches by ascending predicted degree, so it depends on the predictor μ *only through the order it induces* on the resources: two predictors inducing the same order produce identical matchings, and a monotone rescaling of μ changes nothing. The right question is therefore not "how large is the error" but "which *order* error governs the loss, and how tightly." Answering it requires care, because the qualitative fact that order — not magnitude — matters is already a theorem, and we are explicit about what is prior and what is ours.

5.1. What is already known (ACI)

Aamand, Chen and Indyk [1, Appendix D] prove that on the CLV-B model, MinPredictedDegree's matching loss relative to the true expected degrees is at most $n - \mathrm{LIS}(p[\mu])$, where $p[\mu]$ is the true weights ordered by μ and LIS is the longest non-decreasing subsequence — a pure *order* quantity. In particular a monotone (order-preserving) predictor has $p[\mu]$ already sorted, so $n - \mathrm{LIS} = 0$ and the loss is zero. Thus *order-dependence* and *the zero-effect of a monotone bias* are ACI's results, not ours; our `systematic_bias` error model (a monotone rescale) has Kendall-$\tau \equiv 0$ by construction and, consistently,

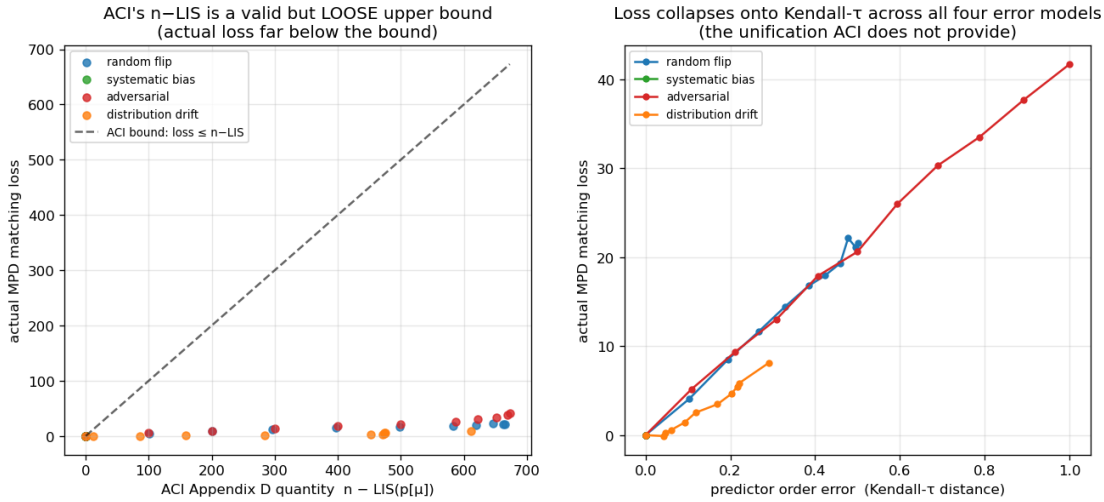


Figure 5.1.: Order error governs the loss: the realized loss lies far below ACI's $n - \mathrm{LIS}$ bound (a) and collapses onto Kendall-τ across all four error models (b).

leaves MPD's ratio exactly unchanged across the benchmark (Chapter 4) — an empirical confirmation of ACI's statement, not a new finding.

5.2. Our characterization

We sweep the four structured error models across strength and record, per model and level on the same instances, three quantities: the actual MPD matching loss, ACI's $n - \mathrm{LIS}$, and the normalized Kendall-τ order error (**Figure 5.1**; $n = 1000$, Zipf exponent 1.0, 40 trials).

(i) ACI's $n - \mathrm{LIS}$ bound is correct but very loose. The realized loss lies far below the bound at every point — by roughly 16× (adversarial) to 75× (distribution-drift); all points hug the axis in Figure 5.1(a).

(ii) $n - \mathrm{LIS}$ saturates and cannot distinguish error structures. For every

non-trivial error it is pinned near its maximum ($\approx$ 610–673 for $n = 1000$), even though the true losses differ by a factor of five across models (8.1 drift, 21.6 random-flip, 41.7 adversarial). As a bound that is almost always $\approx n$, it carries little information about which prediction is more harmful.

(iii) Kendall-τ is the governing order measure, and the models collapse onto it. Plotted against Kendall-τ (Figure 5.1(b)), the four models fall on one increasing curve — loss rises with τ ($0.29 \to 0.50 \to 1.0$ for drift, random-flip, adversarial, tracking $8.1 \to 21.6 \to 41.7$), with `systematic_bias` pinned at $\tau = 0$, zero loss. The quantity that *predicts* the loss and unifies the error structures is the Kendall-τ order distance, which the saturated $n - \mathrm{LIS}$ cannot resolve.

5.3. Chapter summary

We do not claim that order rather than magnitude governs MPD — ACI's Appendix D establishes that, along with the zero-effect of a monotone bias. What we add is a precise empirical characterization: ACI's $n - \mathrm{LIS}$ bound is loose by one to nearly two orders of magnitude and saturates, whereas Kendall-τ predicts the realized loss and unifies the four error structures. This both sharpens the theory's practical content and justifies the single order-error axis used when the predictor is stressed on real data (Chapter 7).

6. Test-and-Fallback in Depth

The test-and-fallback algorithms are the adaptive robustness mechanism of Chapter 4. This chapter gives their first empirical study: the robustness envelope they achieve (§6.1), a counter-intuitive failure of the acceptance threshold (§6.2), its recalibration and the resolution limit that recalibration exposes (§6.3) — a limit that persists across the whole difficulty range and anchors the theoretical outlook of §10.2 — and a benchmark of the dynamic combiner that explains the *commit-once* structure (§6.4). Throughout, the ℓ_1 test is the empirical surrogate the original authors also fall back to (Chapter 3).

6.1. The robustness envelope

Sweeping advice error $\ell_1(p, q)$ on few-types instances ($n = 2000$, $r = 8$, prefix $k = 200$, 40 trials; **Figure 6.1**), FollowPrediction degrades *linearly* from 1.000 at perfect advice to 0.453 at $\ell_1 \approx 1.1$ — well *below* the advice-free floor (≈ 0.99). TestAndMatch instead stays on the **upper envelope**: it captures the benefit when advice is good and, when advice is bad, its prefix test rejects it and it falls back to Ranking, never crashing (BEM $0.998 \to 0.969$; Choo $1.000 \to 0.991$). This is the adaptive counterpart of the structural robustness of Chapter 4.

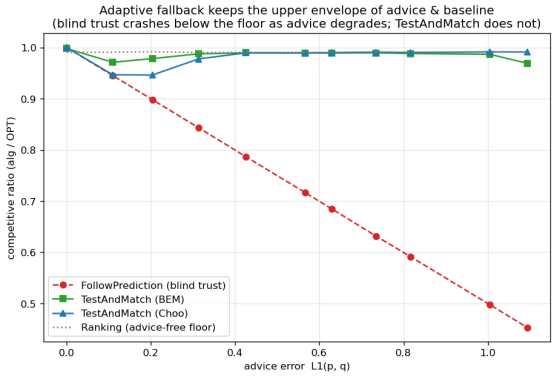


Figure 6.1.: The robustness envelope: blind FollowPrediction crashes below the advice-free floor as advice error grows; TestAndMatch stays on the upper envelope.

6.2. A threshold that is too lenient on average-case inputs

Sweeping the prefix (testing) size k at *borderline* advice ($\eta = 0.15$, true $\ell_1 \approx 0.16$) — where the test works hardest (**Figure 6.2**) — a larger, more accurate test makes the *worse* decision: as k grows $25 \to 800$, the ratio *falls* $0.992 \to 0.956$ and the misjudgement rate *rises* $0.00 \to 0.60$. The mechanism: the Choo/BEM threshold τ is calibrated to the worst-case baseline $\beta \approx 0.696$, but on these instances the baseline is ≈ 0.99 (F3), so the empirical break-even sits at $\ell_1 \approx 0$, far below τ. A small noisy prefix over-estimates ℓ_1 and *accidentally rejects* the borderline advice (landing safely on the floor); a large accurate prefix correctly measures $\ell_1 \approx 0.16 < \tau$ and *accepts* the mildly-bad advice the worst-case threshold deems acceptable — underperforming the baseline. On strong-baseline inputs the worst-case threshold is too lenient, and a more accurate test only follows it more faithfully.

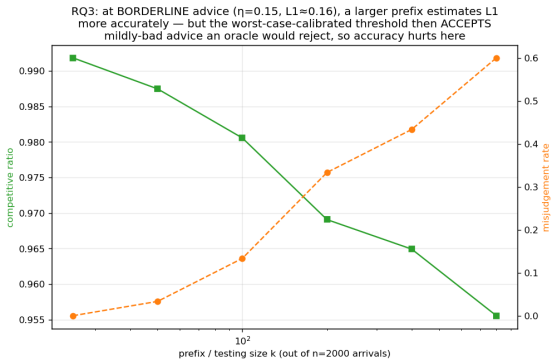


Figure 6.2.: Testing cost at borderline advice: a larger, more accurate prefix test makes the *worse* decision under the worst-case-calibrated threshold.

6.3. Recalibration, and the resolution limit it exposes

Recalibrating τ to the *measured* baseline $\hat{\beta}$ eliminates the pathology (**Figure 6.3**): at borderline advice the worst-case threshold's misjudgement climbs $0.03 \to 1.00$ as the prefix grows and its ratio drops to 0.920, while the recalibrated threshold holds misjudgement 0.00 at every prefix and ratio ≈ 0.986. But recalibration exposes a deeper limit. At perfect advice the worst-case threshold scores 1.000 while the recalibrated one scores only 0.987: the recalibrated $\tau \approx 2(1-\hat{\beta}) \approx 0.028$ is *smaller than the empirical-ℓ_1 estimator's noise floor* ($\approx$ 0.05–0.13), so it can never confidently *accept* — it rejects everything, including perfect advice, and always plays the baseline. In short:

> On strong-baseline instances, no practical empirical-ℓ_1 threshold can both capture the consistency upside and stay safe: the upside is tiny and sits *below the estimator's resolution.* The worst-case threshold over-accepts; the recalibrated one over-rejects; a better tester only follows whichever threshold more faithfully.

Figure 6.4 widens this finding from one threshold to the whole difficulty range.

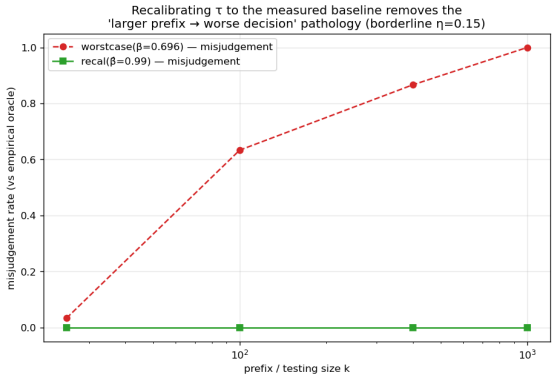


Figure 6.3.: Recalibration removes the threshold pathology: misjudgement holds at zero at every prefix size, versus climbing toward 1.0 under the worst-case threshold.

Sweeping the number of types — the knob that sets the baseline strength — the *potential* upside of perfect advice grows as the baseline weakens, but the upside a sublinear-prefix test can *safely capture* stays pinned near zero: the empirical test's resolution (its noise floor) sits far above the break-even margin wherever an upside exists. The wall of this chapter is therefore not one badly calibrated threshold. The upside and the testing resolution move together — and the concluding outlook (§10.2) gives this coupling a quantitative form: the prefix needed to decide scales as the inverse square of the stakes.

6.4. The dynamic combiner is dominated, and shows why matching needs test-then-commit

We benchmark the Chłędowski-style dynamic combiner to contextualize the commit-once structure of test-and-fallback. In its robust tuning the combiner sits exactly on the floor (0.990 across all advice quality): it never crashes but captures none of the consistency upside, strictly dominated by TestAndMatch. More instructively, an *eager* combiner that switches mid-stream reveals a penalty specific to irrevocable problems: with perfect

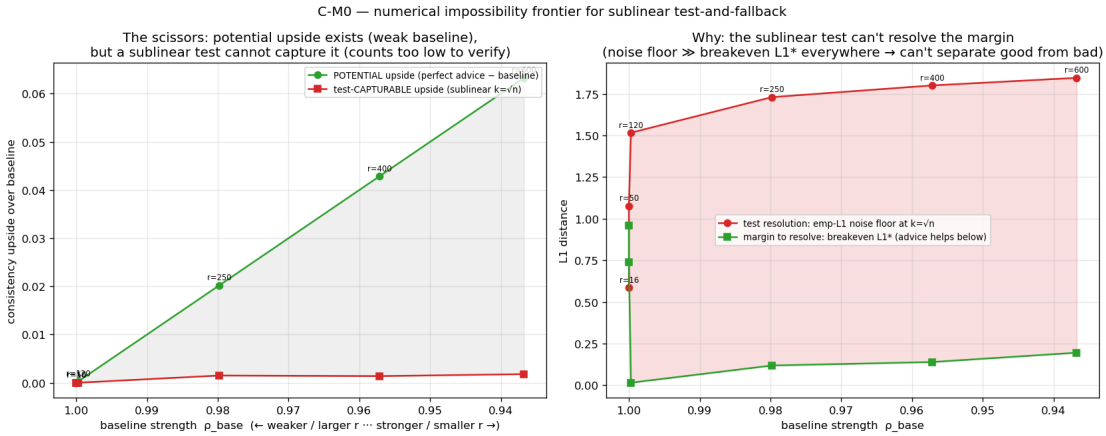


Figure 6.4.: The testing-wall frontier: as the baseline weakens (left), the *potential* upside of perfect advice grows, but the upside a sublinear test can *safely capture* stays near zero — the empirical test's resolution sits far above the break-even margin wherever the upside exists.

advice, eager switching scores 0.927 — *below both* the pure follower (1.000) and the pure baseline (0.958; `tests/test_combiner_small.py`) — because switching from Ranking to advice-following (*mimic*) mid-run lands the committed matching in an *incompatible hybrid*. In an irrevocable problem the follow/fallback decision must be made *before* the bulk of the commitments, which is why Choo/BEM test a prefix and then *commit* rather than switching dynamically. The dynamic combiner that is cheap insurance for caching does not port cleanly to matching.

6.5. Chapter summary

Test-and-fallback delivers the adaptive robustness envelope of §6.1, but its accept/reject decision is fundamentally limited on strong-baseline inputs: no empirical-ℓ_1 threshold

both captures the upside and stays safe (§6.3), and dynamic switching is dominated by test-then-commit (§6.4). The resolution limit of §6.3 — and its persistence across the whole difficulty range (Figure 6.4) — is the empirical anchor of the theoretical outlook in §10.2.

7. External Validity

Chapters 4–6 use synthetic graphs and a synthetic error knob. Is the picture real? We stress it three ways: a genuine, cheap predictor on a real request trace (§7.1); the six real-world graphs (§7.2); and whether *learning* the predictor changes anything (§7.3).

7.1. A real, cheap predictor

We replace the synthetic knob with the cheapest realistic predictor: last-window historical statistics. Real Wikipedia daily pageviews give a live day (the truth) and earlier days (a 1-, 7-, or 30-day-stale forecast), so the error is genuine temporal drift; we map the trace onto a fixed serving topology and consume the forecast through the degree route (MPD) (**Figure 7.1**). Three facts emerge. First, **the cost premise does not bite**: the predictor is a linear-time count (0.108 ms — about 2% of computing OPT once), not an ML inference. Second, **the benefit is real, partial, and never harmful**: a stale forecast captures 27%–68% of the oracle gap (falling with staleness) and always stays above the baseline (0.938–0.957 vs 0.923–0.925, even at 30 days). Third, and why: **topology aggregation makes the cheap predictor order-faithful** — the induced degree predictor's order error is only Kendall-$\tau \approx 0.19$–0.32, roughly *half* the raw histogram's (0.38–0.49) — and since MPD depends only on order (Chapter 5), the aggregated route survives real drift. Consuming the *same* forecast through the raw histogram is catastrophic: blind FollowPrediction collapses to $0.68 \to 0.36$, far below the

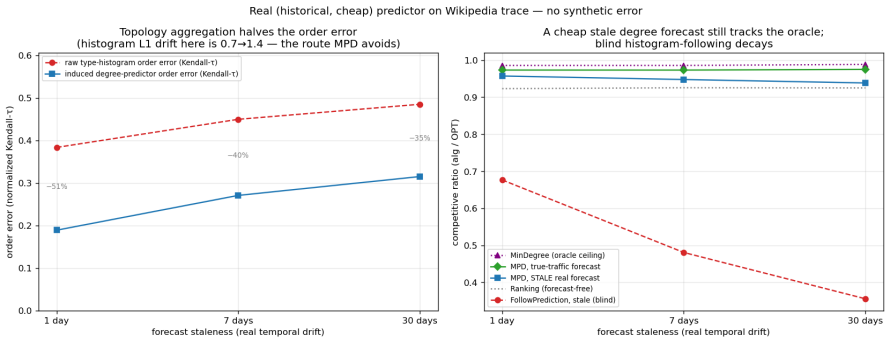


Figure 7.1.: A real, cheap predictor (Wikipedia trace): the aggregated degree route survives staleness (b) because aggregation halves the order error (a); blind histogram-following decays.

≈ 0.92 baseline — exactly what the robust algorithms of Chapters 4 and 6 are for.

7.2. Six real-world graphs

We re-run the degree-prediction roster of Chapter 4 on the six Network-Repository graphs (two Facebook social, two C. elegans biological, two economic input-output), across the same quality columns, with 95% CIs (**Figure 7.2**). The two load-bearing findings are universal. **F1 holds on all six**: naive MPD fed an adversarial predictor falls below the Ranking floor everywhere — by 0.07 (Caltech36) to 0.11 (CE-PG). **F3 is universal and confirms its own logic**: the consistency upside is small everywhere (mean +0.049; range +0.022–+0.077) and smallest exactly where the baseline is strongest — the two dense economic graphs, with Ranking already 0.965/0.977, give the tiniest upsides.

The structural-robustness finding **F2 holds qualitatively on all six** (the augmentations are always less sensitive to prediction quality than naive MPD) and *strictly* on the four social/bio graphs (spread 0.22–0.29× MPD's); on the two dense economic graphs the protection is only *partial* — the augmentation cushions the adversarial drop (0.939

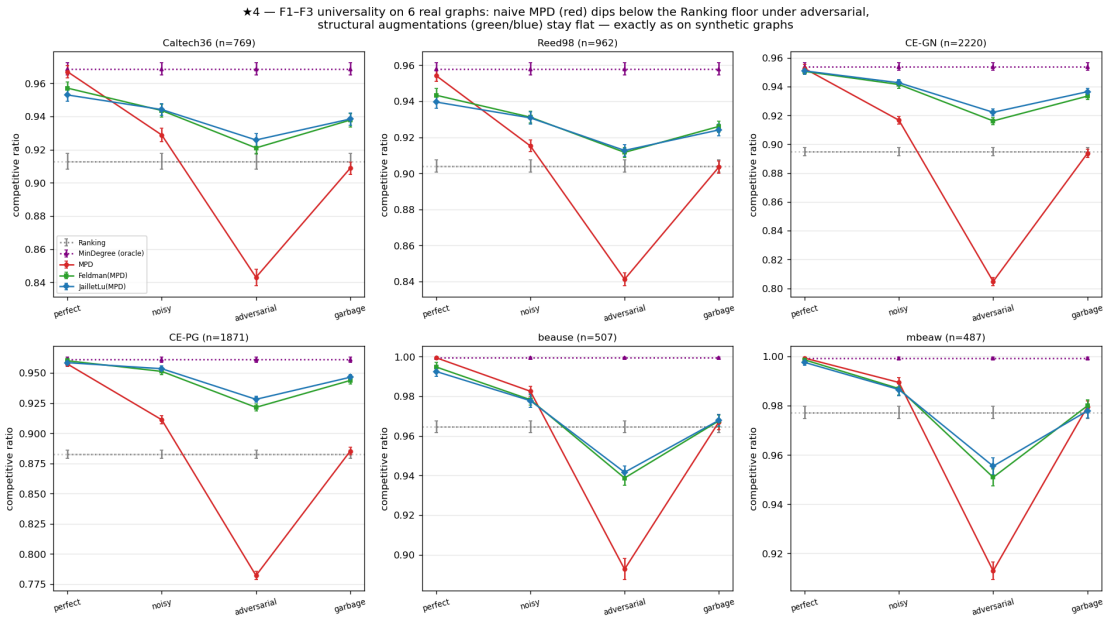


Figure 7.2.: F1–F3 on six real graphs: naive MPD (red) dips below the Ranking floor everywhere; the structural augmentations (green/blue) stay flat.

vs naive MPD's 0.893) but cannot clear the unusually high 0.965 floor, dipping ≈ 0.03 below it. This econ boundary is instructive rather than a failure: those graphs are so dense that matching is nearly trivial (Ranking ≈ 0.97, MinDegree $= 1.00$), so there is neither upside to capture (F3) nor much downside to protect. Finally, F4 is *dramatic*: the worst-case-designed Feldman/Jaillet–Lu are the weakest advice-free entries on the econ graphs (0.73–0.77) and the augmentation lifts them to 0.99 — a +0.26 rescue.

7.3. Does learning the predictor help?

Because MPD consumes the predictor only through order (Chapter 5), one might train it with a rank loss rather than regression. We tested this and report an honest negative.

With *deliberately divergent* synthetic features rank-training beats regression sharply (0.989 ≈ oracle vs 0.974, with worse regression error but better order); but the advantage is gated by feature divergence and by graph difficulty (peaking at +1.3%), and, decisively, on real temporal features it *disappears* — rank- and regression-trained predictors produce identical order (Kendall-τ 0.126 vs 0.126) and identical ratio. The dissociation that powers order-aware training is a property of engineered features, not realistic ones. (The fuller account of this exploration is Chapter 8.) The lesson reinforces the thesis: once a predictor is order-faithful — which a cheap historical count already is (§7.1) — neither a better algorithm nor a better-trained predictor buys much on average-case matching.

7.4. Chapter summary

On real predictors, real graphs, and a learned predictor, the same wall stands: the advice-free baseline is near-optimal, unguarded following is unsafe, and predictions buy downside protection rather than performance. Having established the wall empirically across every setting we could reach, Chapter 8 reports the directions that tried to get past it, and the concluding outlook (§10.2) explains why it is forced.

8. Exploratory Directions and Negative Results

The experimental chapters (4–7) establish a wall: on average-case matching the advice-free baseline is near-optimal, so predictions are robustness insurance rather than a performance lever. Before accepting the wall as final, we pursued three directions that each *tried to get past it* — learning the predictor to squeeze out more performance (§8.1), finding a serving regime where predictions genuinely help (§8.2), and letting a systematic literature review point us to the highest-value contribution (§8.3). All three returned the same answer, and their convergence is what motivated the theorem. This chapter reports them honestly, including the negatives, because they are part of the evidence and because ruling out ambitious alternatives is itself a result.

8.1. Learning to rank the predictor (a negative result)

Chapter 5 shows that MPD consumes its predictor only through the *order* it induces. This suggests a concrete way to do better: rather than training a predictor to minimize its *regression* error to the true degrees (the standard "predict-then-optimize" objective), train it with an *order-aware* loss — a pairwise rank loss — so it optimizes the quantity the algorithm actually uses. We investigated this in three steps.

M0 — the mechanism exists. With deliberately *divergent* synthetic features (one

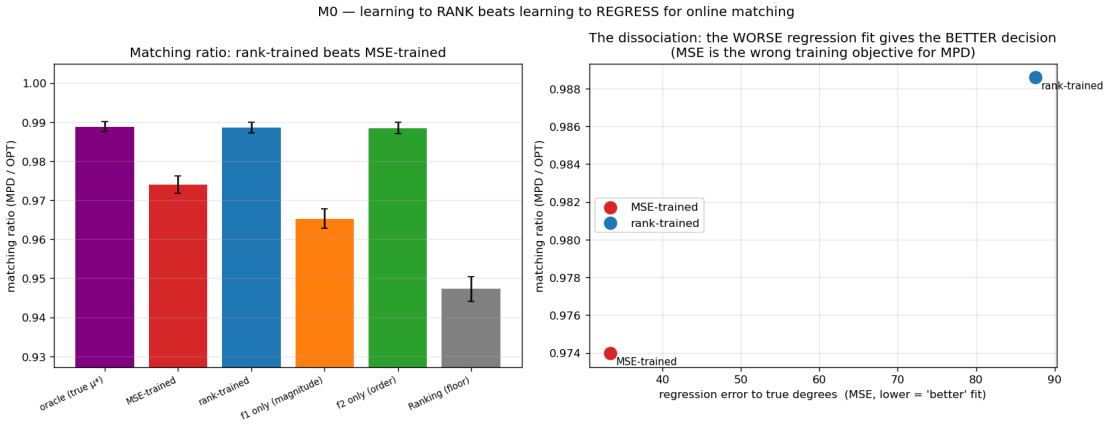


Figure 8.1.: M0: with divergent features, rank-training beats regression on the matching ratio (left) despite a worse regression fit (right).

feature carrying magnitude, another carrying order), a rank-trained linear predictor sharply beats a regression-trained one on the decision metric: matching ratio 0.989 (essentially the oracle) versus 0.974, while the rank-trained predictor has *worse* regression error to the truth (MSE 87.6 vs 33.4) but better order (Kendall-τ 0.058 vs 0.255). The dissociation is real: the worse *fit* gives the better *decision*, confirming that regression is the wrong training objective when it matters.

M1 — but the advantage is doubly gated, and small. Sweeping the feature divergence and the graph difficulty, the rank-advantage is zero when features do not induce an order/magnitude conflict, and zero on easy instances where the baseline is already optimal (the wall, again). It peaks at only +1.3% of the ratio on synthetic graphs; a more favorable framing is *gap-capture*, where rank-training recovers essentially the full oracle gap while regression leaves about 30% of it unrealized — but the gap itself is small.

M3 — and it disappears on real features. The decisive test uses genuine temporal features from real serving traces (per-resource reference counts over the previous windows)

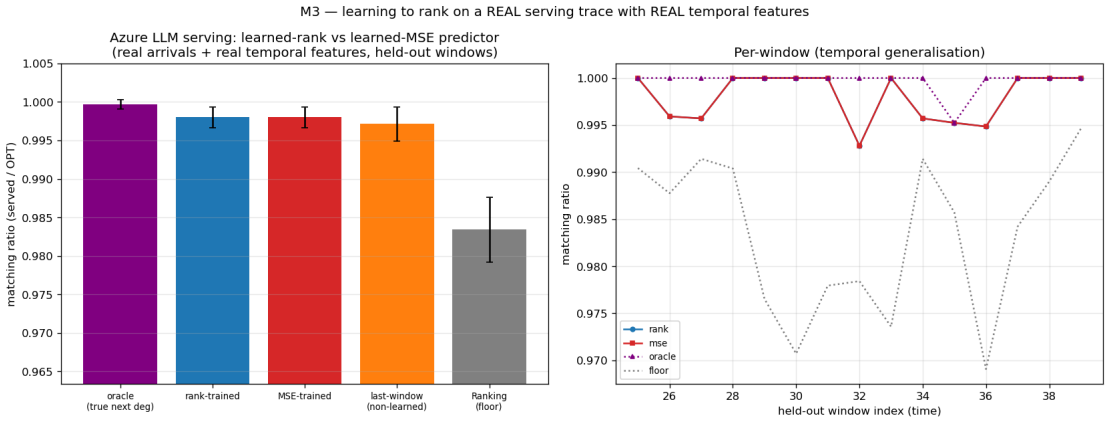


Figure 8.2.: M3 (Azure trace, real temporal features): rank- and MSE-trained predictors are indistinguishable — the engineered divergence that powers rank-training does not arise.

to predict the next window. Here the rank- and regression-trained predictors produce *identical* order (Kendall-τ 0.126 vs 0.126) and identical matching ratio, across every topology and lag configuration tried. The order/magnitude divergence that powers rank-training is a property of *engineered* features; realistic lagged-count features are co-linear noisy estimates of the same popularity, so regression already recovers the order as well as ranking does.

Verdict. Learning the predictor with a decision-aligned loss does not elevate to a standalone contribution: the win requires a feature divergence that does not arise in practice, and even where it does the payoff is bounded by the (small) baseline-to-oracle gap. The result is folded into the thesis as a negative that *reinforces* the central finding — once a predictor is order-faithful, which a cheap historical count already is (Chapter 7), neither a better algorithm nor a better-trained predictor buys much on average-case matching.

8.2. Rescuing the serving application with a with-predictions result (a negative probe)

The serving case study (Chapter 9) recovers established systems results and is therefore presented as a case study rather than a novelty claim. We asked whether a *new* actionable with-predictions result could rescue it. The obstacle is again the wall: every serving variant we tried optimizes *throughput* (goodput), and throughput is forgiving — under overload a reactive router fills capacity just as the optimum does, so predictions add nothing. The escape, if one exists, must be a different *objective* on which the reactive baseline is genuinely far from optimal.

We probed the most promising candidate: an **SLO / tail objective** — protecting a tight-SLO class of requests from being dropped — under bursty, non-stationary load, exactly the regime where a reactive policy, lacking foresight, might fail. Using an event-driven simulator we compared non-predictive policies (static capacity reservation; a reactive-adaptive policy that reserves based on *observed* recent load) against a **clairvoyant oracle** that reserves based on the *actual future* burst. Across every regime swept — overload level, uniform vs bursty tight-SLO demand — the best non-predictive policy matches the clairvoyant oracle to within $\leq 3\%$; in the moderate regime a trivial static reservation of one slot drives tight-SLO violations to near zero and *beats* the clairvoyant oracle outright. Two reasons, both robust to the sweep: protecting a tight-SLO minority needs only a small static headroom, no forecast; and bursts are persistent enough that reacting to observed load is almost as good as forecasting it.

Verdict. The tail objective is forgiving too — a third face of the wall, after throughput (Chapters 4–7) and predictor-learning (§8.1). We found no natural regime where foresight helps, so serving remains a case study. A regime that would break the wall (a non-stationary or adversarial objective where the baseline is far from optimal) is exactly the kind of setting the thesis brackets as future work.

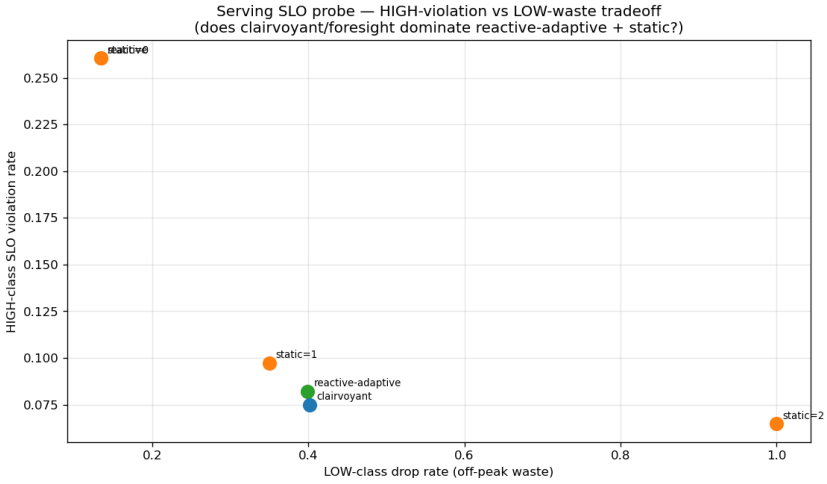


Figure 8.3.: Serving SLO probe: a non-predictive policy matches the clairvoyant oracle to within a few percent — foresight does not help.

8.3. A literature review that redirected the work

Deciding *which* of our findings were genuinely novel — and worth developing — required a systematic prior-art review rather than intuition. We conducted a large multi-source, adversarially-verified literature search (documented in `docs/LITERATURE_REVIEW.md`) that returned honest, sometimes deflating, verdicts. It confirmed that the unified experimental benchmark and the empirical study of test-and-fallback are unoccupied and worth leading with; that the order-error finding is *partially* pre-empted by ACI's Appendix D and must be reframed as a tightness/measure characterization rather than a discovery (Chapter 5); and that the serving results are largely re-derivations of established systems facts, warranting their demotion to a case study (§8.2, Chapter 9). A second, focused prior-art pass on the theory question confirmed that no prior work quantifies the cost of the prefix test on strong-baseline instances, while flagging the one risk any such result must defend against (Choo et al.'s constructive baseline-coupling) — both inputs

to the outlook of §10.2 and to the companion theoretical work it defers to.

This review is itself part of the research process reported here: it turned an undifferentiated pile of findings into a prioritized contribution, redirected effort away from low-value elaborations (weighted-value variants, more baseline comparisons) and toward the theory question, and enforced the honesty guardrails carried throughout the thesis.

8.4. Synthesis: the negatives point to the same wall

The three explorations point the same way. A better-trained predictor does not help on real features (§8.1); a with-predictions lens does not rescue serving even on a tail objective (§8.2); and the literature review confirmed there was no easy performance win to be had (§8.3). Together with the throughput wall of Chapters 4–7, they show the phenomenon is not confined to one objective, one algorithm, or one dataset — it recurs everywhere we pushed. That robustness of the empirical finding is precisely what suggested it might be *forced* rather than accidental — the question the concluding outlook (§10.2) takes up, and which the companion theoretical work in preparation pursues in full.

9. Application Case Study: AI-Inference Serving

The thesis's abstraction is not confined to classical matching markets; it instantiates a contemporary systems problem. This chapter casts request routing in an AI-inference serving system as online matching and shows the framework recovers the field's established results. It is presented, deliberately, as a **case study, not a novelty claim** — the systems facts below are known, and the with-predictions vocabulary is a re-labeling of them (a decision justified by the literature review of Chapter 8 and the negative probe summarized in §9.2).

9.1. The serving instantiation

We map serving to online b-matching: the offline resources are model replicas or cache shards, each with a **capacity** c (it can serve up to c concurrent requests, hence b-matching rather than 1-matching); arrivals are requests drawn from a non-uniform, bursty traffic distribution over request types; an edge is a capability or cache affinity; and **goodput** — the fraction of requests served — is the competitive ratio against the b-matching optimum. We instantiate this on three real traces (Wikipedia pageviews, an Azure LLM inference trace, and the Mooncake prefix-cache trace [15]) and across four serving concerns.

Capacity as robustness (**Figure 9.1**). Increasing the capacity c smooths the effect

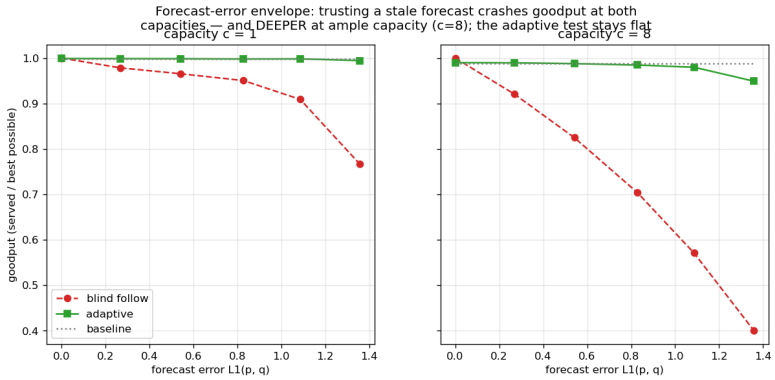


Figure 9.1.: Capacity as robustness: blindly following the forecast crashes goodput — deeper at ample capacity ($c = 8$) — while the adaptive test stays flat.

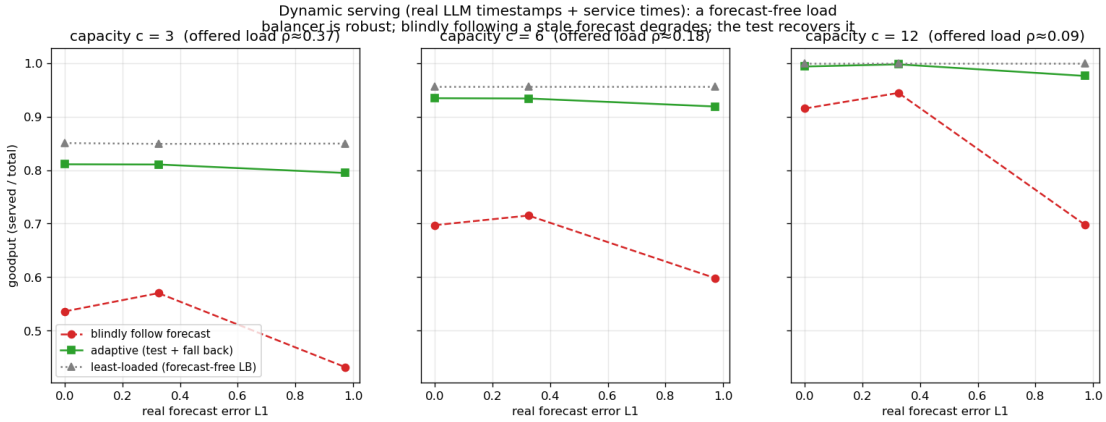


Figure 9.2.: Live load beats a stale forecast under dynamic service times; an adaptive test recovers most of the gap.

of a bad prediction: a capacity-aware baseline stays safe, while *blindly* trusting a traffic forecast degrades *further* as capacity grows. Capacity is thus a *substitute* for algorithmic robustness — the systems analogue of the robustness-insurance thesis.

Forecasts vs live load (**Figure 9.2**). Under dynamic service times (requests hold a

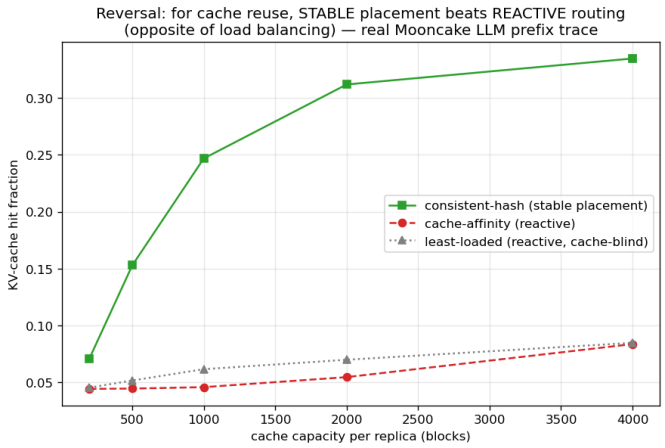


Figure 9.3.: The cache-affinity reversal (Mooncake trace): stable placement beats reactive routing for KV-cache reuse.

slot for a real duration, released event-by-event), a live-load signal beats a stale traffic forecast — a reactive load balancer outperforms a forecast-following router when the forecast has aged.

Cache-affinity routing (**Figure 9.3**). For prefix-cache-aware routing, a *stable* affinity router beats a reactive one — the reverse of the traffic-forecast case, because cache locality rewards persistence [17].

Each of these recovers an established systems result cleanly, demonstrating that the framework instantiates the problem faithfully.

9.2. A probe for a genuinely new result, and its negative

Because the literature suggested the with-predictions lens might yield a *new* actionable serving result on a tail objective, we probed it (the full account is in Chapter 8). On an SLO/tail objective — protecting a tight-SLO class of requests under bursty load — a non-predictive policy (static headroom, or a reactive-adaptive reservation based on

observed load) matches a clairvoyant oracle to within $\leq 3\%$ across every regime swept; a trivial static reservation of one slot even beats the oracle in the moderate regime. Two reasons, both robust: protecting a tight-SLO minority needs only a small static headroom, no forecast; and bursts are persistent enough that reacting to observed load is nearly as good as forecasting it. The tail objective is forgiving too — a third face of the thesis's wall, after throughput (Chapters 4–7) and predictor-learning (Chapter 8). We found no natural regime where foresight helps, so serving remains a case study.

9.3. Chapter summary

The serving instantiation shows the framework reaches a live systems problem and recovers its established results — capacity as a robustness substitute, live load over stale forecasts, stability for cache locality — and that even a tail objective does not open a genuine with-predictions win. The application is a faithful case study of the thesis, not an independent contribution.

10. Conclusion and Future Work

10.1. Summary

This thesis studied learning-augmented algorithms for online bipartite matching by first building a common experimental foundation and then explaining what it revealed. On a single harness — validated against the published Borodin et al. study (Chapter 3) — we compared the advice-free baselines, MinPredictedDegree and its augmentations, and the test-and-fallback algorithms, across synthetic graphs, six real-world graphs, and real request traces (Chapters 4–7). Across every setting the same picture appeared: the advice-free baseline is already near-optimal on average-case inputs, so the *consistency* upside of a good prediction is small; unguarded prediction-following crashes *below* the baseline under bad predictions; and the value of the sophisticated algorithms is downside protection, delivered by either a structural or an adaptive robustness mechanism with distinct trade-offs. We sharpened the sub-questions — that the (small) loss is governed by a Kendall-τ order error rather than magnitude (Chapter 5, crediting the prior order-dependence result of ACI), and that the adaptive test has a threshold-calibration pathology and a resolution limit (Chapter 6) — and we honestly reported the directions that did *not* pan out: learning the predictor for extra performance, and finding a serving regime where predictions genuinely help (Chapter 8).

The recurring wall raises an obvious question: is it an accident of the inputs we chose, or is it forced? The next section gives our answer in outlook form. The single

sentence that unifies the thesis is: **on average-case online matching, predictions are robustness insurance rather than a performance lever — and the upside they offer is smaller than the price of finding out whether to trust them.**

10.2. A theoretical outlook: the price of testing advice

The experiments say that no *practical* acceptance threshold captures the upside safely (§6.3, Figure 6.4). This section sketches, without claiming a full theorem, why no decision rule of *any* kind escapes the wall on the inputs where it stands. One ingredient is proved here; the rest is a quantitative reading of our own experiments, whose full formal development is deliberately deferred to companion work in preparation.

> **A trade-off inequality.** Let G and Bd be two instance distributions sharing the *same* advice, such that following the advice gains δ under G and loses Δ under Bd relative to the advice-free baseline, and let γ_k be the total-variation distance between the laws of their length-k prefixes. Then every test-and-fallback algorithm — deciding by *any* measurable rule on its prefix — satisfies
>
> $$(1 - \eta_c) \;\leq\; \eta_r + \gamma_k + o(1),$$
>
> where η_c is the fraction of the upside it forgoes under G and η_r its robustness loss under Bd as a fraction of Δ.

The proof is a short conditioning argument: capturing the upside forces the algorithm to follow with high probability under G; robustness forces fallback under Bd; and no function of the prefix can behave differently on two prefix distributions that are statistically γ_k-close. The inequality thus converts "consistent *and* robust" into a question about *sample complexity*: how long must the prefix be before it distinguishes advice worth following from advice worth rejecting?

The answer, on the rare-resource instances that produce Figure 6.4, is a **budget–stakes law**: the decision costs a prefix of $k^* = \tilde{\Theta}(\theta/\delta^2)$ — the inverse square of the stakes δ (what good advice gains over the baseline), scaled by the contention θ — and the budget is achievable, by a simple *directional* statistic (do the prefix arrivals agree with the advice's predictions more often than they contradict them?). The stakes, in turn, are capped by the baseline slack: $\delta \leq 2\varepsilon(1 - \rho_{\text{base}})$. Substituting, on strong-baseline instances the required prefix exceeds the entire instance — any upside below $\Theta\big(\sqrt{(1 - \rho_{\text{base}})/n}\big)$ cannot be captured safely by any rule at any prefix length $k \leq n$. At the parameters of Chapter 4 ($\rho_{\text{base}} \approx 0.99$, $n = 2000$) that threshold is ≈ 0.004, and the upsides we measured are of exactly this order (F3): the empirical wall sits in the regime the budget law governs. This is Figure 6.4 read quantitatively: the potential upside and the capturable upside separate as the baseline strengthens, because the stakes shrink faster than the test's resolution improves.

Two honest notes bound the scope of this outlook. First, the budget law cuts both ways: where the stakes are large (a weak baseline), the directional statistic is cheap and following good advice is easy — the wall is a statement about strong-baseline, average-case inputs, not about testing in general. In particular, a tempting stronger conjecture — that the near-linear sample cost of *tolerant distribution testing* (§2.5) blocks every sublinear rule outright — is false: it is refuted by the same directional statistic, and the ℓ_1-threshold blindness of §6.3 is a property of testing the *distance* rather than the *payoff*. Second, this thesis claims here only the inequality displayed above and the quantitative reading of its own experiments; the two-sided law, its proofs, and its exact scope are the subject of the companion work.

10.3. Limitations

- **Input model.** We work in the known-i.i.d. model. Because Known-I.I.D. $\leq$ Random-Order in difficulty, the algorithms' guarantees carry over, but the empirical wall is an *average-case* statement; we do not claim it for adversarial arrival order.
- **Test model.** Following the original authors, the test-and-fallback experiments use an empirical-ℓ_1 surrogate for the (unimplemented) distribution tester; §10.2 explains why the surrogate's blindness is structural rather than an implementation artifact.
- **Prediction-object heterogeneity.** The degree- and histogram-prediction families do not map onto every graph, which is why they are reported in parallel panels rather than one table.
- **Data breadth.** Each real modality is exercised by one trace.
- **Theory scope.** The thesis deliberately confines its theory to the outlook of §10.2 — one proved trade-off inequality and a quantitative reading of the experiments. The full budget–stakes law is companion work in preparation, and no theorem beyond the stated inequality is claimed here.

10.4. Future work

The thesis brackets, rather than resolves, the settings in which predictions might genuinely help online matching, and these are the natural next directions.

- **Beyond average-case inputs.** The wall is an average-case phenomenon. Adversarial or non-stationary arrival orders, where the advice-free baseline is provably far from optimal, are where predictions should carry real value — and where a *positive* counterpart to this thesis's wall might be proved.
- **Beyond throughput.** Objectives on which the baseline is not near-optimal —

tail latency, per-type fairness, migration or recompute cost — may admit genuine with-predictions gains that the goodput objective forecloses; our serving SLO probe (Chapter 8) closed the simplest such attempt but not the space.

- **Completing the theory.** Finishing the companion budget–stakes development — the sharp two-sided law, the directional statistic as its matching upper bound, and its exact scope (decomposable families; whether a non-decomposable family can push the budget higher is open) — would turn the outlook of §10.2 into a tight characterization.
- **Better tests, honestly.** Whether a super-linear or amortized test, or a test that reuses online decisions as samples, can beat the stakes-squared budget of §10.2 is an open and practically motivated question.

10.5. Closing

Predictions are a powerful tool for online algorithms, but this thesis is a study of their *limits* on one well-understood problem. On average-case online matching, the honest verdict is that a cheap, order-faithful predictor already captures nearly all there is to capture, that the sophisticated machinery earns its keep as insurance rather than as performance, and that finding out whether to trust a prediction costs more, on these inputs, than the prediction is worth. Recognizing where predictions cannot help is, we hope, as useful as knowing where they can.

A. Reproduction Guide

Every quantitative result in this thesis is regenerated from a fixed seed by a single script. This appendix maps each figure and table to its script, gives the commands and runtimes, lists the full Phase-2 reproduction tables deferred from Chapter 3, and records data provenance.

A.1. Environment and principles

- **Stack:** Python 3.12; NumPy 1.26+, SciPy 1.13, NetworkX 3.3, Matplotlib (plots only).
- **Determinism:** all randomness derives from one master seed (default `0`) via NumPy's `default_rng(seed).spawn(k)`, which yields independent, reproducible streams (graph, instance, algorithm randomness, prediction perturbation). Results are bit-stable across NumPy minor versions from 1.25 (BitGenerator-based spawning).
- **Paired trials:** within a comparison, every algorithm and error level reuses the same graph, arrival sequence, `OPT`, and tie-break seed, so differences are attributable to the prediction alone.
- **Confidence intervals:** 95% normal-approximation half-widths over trials.
- Each script writes `results/<name>.json` (raw means/CIs) and `results/<name>.png` (plot), and prints per-step progress.

A.2. Figure / table → script map

Thesis object	Script	Output	≈ runtime
Fig 3.1 (ER U-curve)	`scripts/run_er_full.py`	`results/er_full.{json,png}`	~20 min
Fig 3.2 (left-regular)	`scripts/run_left_regular.py`	`results/left_regular.{json,png}`	~10 min
Table 4.1 (unified benchmark)	`scripts/run_unified_benchmark.py`, then `plot_unified_panels.py`	`results/unified_benchmark.{json,png}`, `unified_benchmark_panel{A,B,C}.png`, `unified_benchmark_tables.md`	~100 s
Fig 4.1 (consistency–robustness plane)	`scripts/run_consistency_vs_robustness.py`	`results/consistency_vs_robustness.{json,png}`	~1 h
Fig 5.1 (order-error vs ACI)	`scripts/run_order_vs_theory.py`	`results/order_vs_theory.{json,png}`	~30 s
Fig 6.1 (envelope), Fig 6.2 (prefix sweep)	`scripts/run_choo_bem.py`	`results/choo_bem_{envelope,prefix}.png`	~20 min
Fig 6.3, recalibration (§6.3)	`scripts/run_recalibration.py`	`results/recalibration_*.png`	~1.5 min
Fig 7.1 (real predictor)	`scripts/run_real_predictors.py`	`results/real_predictors.{json,png}`	~15 s
Fig 7.2 (six real graphs)	`scripts/run_real_robustness.py`	`results/real_robustness.{json,png}`	~65 s
Fig 8.1 (M0 rank vs MSE)	`scripts/run_rank_vs_mse.py`	`results/rank_vs_mse.{json,png}`	~10 s
M1 sweep (§8.1, no figure)	`scripts/run_rank_when_it_matters.py`	`results/rank_when_it_matters.{json,png}`	~20 s

Thesis object	Script	Output	≈ runtime
Fig 8.2 (M3 real-trace learning)	`scripts/run_rank_real_trace.py`	`results/rank_real_trace.{json,png}`	~10 s
Fig 8.3 (serving SLO probe)	`scripts/run_serving_slo_probe.py`	`results/serving_slo_probe.{json,png}`	~1 s
Fig 6.4 (testing-wall frontier)	`scripts/run_impossibility_frontier.py`	`results/impossibility_frontier.{json,png}`	~1 s
Figs 9.1–9.3, serving (Ch 9)	`scripts/run_serving.py`, `run_serving_trace.py`, `run_serving_dynamic.py`, `run_prefix_cache.py`	`results/serving_*.png`, `prefix_cache_*.png`	~1 s each
Real-world Borodin Tables 3/4 (validation)	`scripts/run_realworld.py`	`results/realworld.json`	~few min

A.3. Reproduction commands

```
# from the project root (matching-experiments/)
# correctness anchors - 7 hand-verifiable test files, all pass:
for t in tests/test_*.py; do python3 "$t"; done

# regenerate the headline results (fast ones):
python3 scripts/run_unified_benchmark.py        # Table 4.1 (data)
python3 scripts/plot_unified_panels.py          # Table 4.1 (panel charts)
python3 scripts/run_order_vs_theory.py          # Fig 5.1
python3 scripts/run_real_predictor.py           # Fig 7.1
```

```
python3 scripts/run_realworld_robustness.py     # Fig 7.2
python3 scripts/run_impossibility_frontier.py   # Fig 6.4
python3 scripts/run_rank_vs_mse_mve.py          # Fig 8.1
python3 scripts/run_rank_when_it_matters.py     # M1 (§8.1)
python3 scripts/run_rank_real_trace.py          # Fig 8.2
python3 scripts/run_serving_slo_probe.py        # Fig 8.3
python3 scripts/run_consistency_robustness.py   # Fig 4.1 (replots Table 4.1)

# the long (max-flow / Hopcroft-Karp-bound) sweeps:
python3 scripts/run_er_full.py                  # Fig 3.1  (~20 min)
python3 scripts/run_left_regular.py             # Fig 3.2  (~10 min)
python3 scripts/run_choo_bem.py                 # Fig 6.1, 6.2 (~20 min)
```

All scripts hardcode seed `0` unless noted; re-running reproduces the reported numbers.

A.4. Full Phase-2 reproduction tables (deferred from §3.6)

Setup: $n = 1000$, $m = n$, 100 trials per parameter value, seed 0. Target is *qualitative* agreement with Borodin et al. (Python/NetworkX vs the paper's C++/Edmonds–Karp; absolute differences ≤ 0.02 accepted).

Erdős–Rényi, competitive ratio at selected c (SG=SimpleGreedy, Rk=Ranking, F/J = Feldman/Jaillet–Lu, -NG/-G = non-greedy/greedy):

c	SG	Rk	F-NG	F-G	J-NG	J-G
0.10	0.9995	0.9994	1.0000	1.0000	0.9991	1.0000
1.90	0.9362	0.9363	0.9033	0.9637	0.9202	0.9602
4.90	**0.8640**	0.8655	0.7648	**0.8845**	0.7949	**0.8854**

c	SG	Rk	F-NG	F-G	J-NG	J-G
8.90	0.9094	0.9088	0.7314	0.9123	0.7662	0.9120
14.90	0.9487	0.9486	0.7290	0.9488	0.7640	0.9483

Per-algorithm minima (ER): SG 0.8640 @ c=4.9; Rk 0.8649 @ 4.7; F-NG 0.7288 @ 13.5; F-G 0.8833 @ 5.3; J-NG 0.7616 @ 14.1; J-G 0.8839 @ 5.3.

Random left-regular, competitive ratio at selected d:

d	SG	Rk	F-NG	F-G	J-NG	J-G
1	1.0000	1.0000	0.9845	1.0000	0.9845	1.0000
2	0.9539	0.9537	0.8769	0.9679	0.8945	0.9659
5	**0.8905**	0.8900	0.7595	0.9008	0.7909	0.9022
10	0.9275	0.9278	0.7317	0.9276	0.7677	0.9290
30	0.9770	0.9767	0.7308	0.9757	0.7633	0.9754

Paper-claim checklist (all verified ✓): greedy minima near $c \approx 4.9$ / $d = 5$; Ranking ≈ SimpleGreedy (max diff 0.0017 ER, 0.005 LR — the paper omits Ranking's curve for this reason); non-greedy variants degrade monotonically as c, d grow; greedy complex variants ≈ SimpleGreedy asymptotically; the c=14.9 non-greedy ordering (J-NG 0.764 > F-NG 0.729) matches the paper's worst-case-bound ordering. A cross-family observation: the non-greedy algorithms converge to the same asymptotic constants in both families (0.729 Feldman, 0.764 Jaillet–Lu, within 0.002), above their worst-case bounds by +0.06 / +0.03 — an early instance of the average case being more benign than the worst case.

A.5. Data provenance

Real data is stored locally under `data/` (large; excluded from version control).

- **Real graphs (Chapter 7, §3.6 validation):** six Network Repository graphs — `socfb-Caltech36`, `socfb-Reed98`, `bio-CE-GN`, `bio-CE-PG`, `econ-beause`, `econ-mbeaflw` — as MatrixMarket `.mtx` / whitespace `.edges`, reduced to simple undirected graphs and converted to bipartite by random balanced partition (Borodin Table 3) or duplicating double-cover (Table 4).
- **Traces (Chapters 7, 8, 9):** Wikipedia "top articles per day" JSON for four days (`data/trace/wiki/`, used as live day vs 1/7/30-day-stale forecasts); the Azure LLM inference trace (`data/trace/azure_llm/`, context/generated token counts with timestamps); the Mooncake conversation trace (`data/trace/mooncake/`, per-request `hash_ids` for prefix-cache blocks).

A.6. Outlook verification snippets (§10.2)

The quantities behind the outlook of §10.2 are numerically verified. The single-cell constants of the rare-resource construction (per-cell OPT, baseline, and the advantage/ℓ_1 formulas) and the exact affine conversion law $\mathbb{E}[\text{follow-ratio}] = \rho_{\text{perfect}} - \frac{1}{2}\ell_1(p, q)$ were checked to three decimals by short simulations documented in the project notes (`docs/T1_W1_single_cell.m` `T1_W2_W3a_closeout.md`); e.g. at $\theta = 0.6$, bias $|s - \frac{1}{2}| = 0.3$, the simulated per-cell advantage ± 0.119 matches the formula $\pm\theta|s - \frac{1}{2}| = \pm 0.12$. The budget–stakes claims — the directional statistic's accuracy at fixed prefix length independent of n, and the blindness of the plug-in ℓ_1 estimate at $k \ll r$ — are verified by `scripts/verify_witness_gap.py` (`docs/T1_WITNESS_GAP.md`). These support the outlook; the formal development is companion work outside this thesis.

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